

NTC_CC105 – DATA STRUCTURE W/LAB

WEEK 5 LABORATORY INSTRUCTION

Laboratory Activity 5

Instructor	Prof. Darwin C. Vargas, MIT
Program	BSIT
Academic Year	2026–2027
Programming Language	C++

1. Student Name	Quintela, Nicole M. / Serdenia Rihanna Shealya
2. Student Number	425001249 / 425002643
3. Section	2.4 BSIT
4. Date	Oct 5, 2026
5. Laboratory Activity No.	Laboratory Activity 5
6. Laboratory Title	Linear Architecture Synthesis & Peer Code Audit

PART E: Peer Review Evaluation Form

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CC105 PEER CODE AUDIT EVALUATION FORM

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Author Name : [Quintela,Nicole M. / Serdenia, Sheayla Rihanna A.]

Reviewer Name : [Quintela,Nicole M. / Serdenia, Sheayla Rihanna A.]

Date of Audit : [October 5, 2026]

Repository Link: [<https://github.com/425002643-debug/Laboratory-Activity>]

1. IDENTIFIED STRENGTHS (At least 3 positive engineering practices observed)

[1] Properly applied RAI patterns by handling memory cleanup using delete[] inside the destructor.

[2] Managed out-of-bounds array access effectively through exceptions and error-indicating boolean returns.

[3] Eliminated hardcoded magic numbers in favour of descriptive const variables and explicit templates.

2. IDENTIFIED WEAKNESSES / DEFECTS (At least 3 structural or style issues)

[1] The processMatrix() function still uses static parameter sizing.

[2] Lacks a copy constructor and copy assignment operator (Rule of Three violation).

[3] Missing explicit inline comments explaining the matrix transposition loop.

3. RECOMMENDED ACTIONABLE IMPROVEMENTS (At least 3 recommendations)

[1] Implement dynamic template support <typename T> for generic data types.

[2] Add an explicit shrink_to_fit() method to optimize unused allocated memory.

[3] Mark getter functions (e.g., getSize(), Search()) with the `const` modifier.

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DEVELOPER ACTION & REJECTION JUSTIFICATION (To be filled by Author)

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Recommendation [1]: [ACCEPTED] - Will refactor dynamic array to use template classes.

Recommendation [2]: [PARTIALLY ACCEPTED] - Kept manual capacity controls for Milestone 1 scope.

Recommendation [3]: [ACCEPTED] - Added `const` specifiers to all accessor methods.

Author Signature: Nicole Quintela / Sheayla Serdenia

Date: October 5, 2026